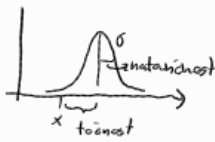


Kovarianca / korelacija

merimo neznano količino X

$$z_1 \sim \mathcal{N}(x, \sigma_1^2) \text{ (normalna porazdelitev)} \rightarrow z_1 = x + r_1, \quad r_1 \sim \mathcal{N}(0, \sigma_1^2) \quad \sigma_1^2 = \langle (z_1 - x)^2 \rangle = \langle r_1^2 \rangle$$

$$z_2 \sim \mathcal{N}(x, \sigma_2^2) \rightarrow z_2 = x + r_2, \quad r_2 \sim \mathcal{N}(0, \sigma_2^2)$$



če imamo odvisne meritve: $r_2 = \alpha r_1 + w$, $w \sim \mathcal{N}(0, \sigma_w^2)$

$$\sigma_2^2 = \langle r_2^2 \rangle = \dots = \alpha^2 \sigma_1^2 + \sigma_w^2 + 2\alpha \langle r_1 w \rangle$$

$\langle r_1 w \rangle \dots$ kovarianca

$\langle r_1 w \rangle = 0$, ker nista odvisna

$$1 = \left(\alpha \frac{\sigma_1}{\sigma_2} \right)^2 + \left(\frac{\sigma_w}{\sigma_2} \right)^2$$

$$0 \leq \left(\alpha \frac{\sigma_1}{\sigma_2} \right)^2 \leq 1 \rightarrow -1 \leq \rho_{12} \leq 1$$

ρ_{12} korelacijski koeficient

$\rho_{12} = 0 \Rightarrow \alpha = 0$ z_1, z_2 neodvisni spr.

$\rho_{12} = \pm 1 \rightarrow$ popolna (anti)korelacija

$$\text{kovarianca: } \sigma_{12} = \langle (z_1 - x)(z_2 - x) \rangle = \langle r_1 r_2 \rangle = \langle r_1 (\alpha r_1 + w) \rangle = \alpha \sigma_1^2$$

$$\rightarrow \rho_{12} = \alpha \frac{\sigma_1}{\sigma_2} / \sigma_1 \sigma_2$$

$$\alpha \sigma_1^2 = \sigma_{12} = \rho_{12} \sigma_1 \sigma_2$$

Širjenje napak (splošni predpis)

$$\left. \begin{aligned} u_1 &= f_1(x_1, \dots, x_n) \\ u_2 &= f_2(x_1, \dots, x_n) \\ &\vdots \\ u_k &= f_k(x_1, \dots, x_n) \end{aligned} \right\} \text{ uvedemo vektorski zapis: } \vec{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \quad \vec{u} = \begin{pmatrix} u_1 \\ \vdots \\ u_k \end{pmatrix}$$

$$\hookrightarrow \text{poznamo oče } \vec{x} = \begin{pmatrix} \bar{x}_1 \\ \vdots \\ \bar{x}_n \end{pmatrix}$$

$$\rightarrow \text{def. kov. } \vec{x}_i = x_i + m_i, \quad m_i \sim \mathcal{N}(0, \sigma_i^2)$$

$$\sigma_{ij} = \langle (\bar{x}_i - x_i)(\bar{x}_j - x_j) \rangle = \langle m_i m_j \rangle$$

$$\sigma_{ii} = \langle m_i^2 \rangle = \sigma_i^2 \text{ varianca}$$

$\rightarrow$ uvedemo kovariančno matriko:

$$M_{ij} = \sigma_{ij} = \langle (\bar{x}_i - x_i)(\bar{x}_j - x_j) \rangle = \langle m_i m_j \rangle$$

$$M = \langle (\vec{\bar{x}} - \vec{x})(\vec{\bar{x}} - \vec{x})^T \rangle = \langle \vec{m} \vec{m}^T \rangle$$

- simetrična $M = M^T$
- diagonalni elementi: $M_{ii} = \sigma_i^2$ variance
- izvan-diagonalni elementi $M_{ij} = \sigma_{ij}$ kovariance

$$\vec{u} = ? \quad U = ? \text{ kov. matr.}$$

$$\text{Taylor: } \vec{u} = \begin{pmatrix} f_1(\vec{x}) \\ \vdots \\ f_k(\vec{x}) \end{pmatrix} + J_{\vec{u}}(\vec{x})(\vec{x} - \vec{\bar{x}}) + \sigma(\vec{x})$$

$$\vec{u} = \begin{pmatrix} f_1(\vec{x}) \\ \vdots \\ f_k(\vec{x}) \end{pmatrix} + J_{\vec{u}}(\vec{x})(\vec{x} - \vec{\bar{x}})$$

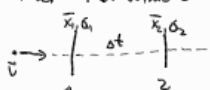
$$\rightarrow \vec{u} - \vec{u} = \vec{u} - (J_{\vec{u}}(\vec{x})(\vec{x} - \vec{\bar{x}})) = J_{\vec{u}}(\vec{x})(\vec{\bar{x}} - \vec{x})$$

$$\rightarrow U = \langle (\vec{u} - \vec{u})(\vec{u} - \vec{u})^T \rangle =$$

$$= \langle J_{\vec{u}}(\vec{x})(\vec{\bar{x}} - \vec{x})(\vec{\bar{x}} - \vec{x})^T J_{\vec{u}}^T(\vec{x}) \rangle =$$

$$= J_{\vec{u}}(\vec{x}) M J_{\vec{u}}^T(\vec{x})$$

Primer: TOF (time of flight)



$\bar{x}_1, \bar{x}_2$ neodvisni

Izračunaj kov. matriko za $\vec{u} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$

$$\vec{x} = \begin{pmatrix} \bar{x}_1 \\ \bar{x}_2 \end{pmatrix} \rightarrow M = \begin{pmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{pmatrix} \quad \text{ker so } \bar{x}_1, \bar{x}_2 \text{ neodvisni}$$

$$\vec{u} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} f_1(\vec{x}) \\ f_2(\vec{x}) \end{pmatrix}$$

$$v = f_3(\vec{x}) = \frac{\bar{x}_1 - \bar{x}_2}{\Delta t}$$

$$J_{\vec{u}} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -\frac{1}{\Delta t} & \frac{1}{\Delta t} \end{bmatrix}$$

$$\rightarrow U = J M J^T =$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -\frac{1}{\Delta t} & \frac{1}{\Delta t} \end{bmatrix} \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix} \begin{bmatrix} 1 & 0 & -\frac{1}{\Delta t} \\ 0 & 1 & \frac{1}{\Delta t} \end{bmatrix} =$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -\frac{1}{\Delta t} & \frac{1}{\Delta t} & \frac{1}{\Delta t^2} \end{bmatrix} \begin{bmatrix} \sigma_1^2 & 0 & \sigma_1^2 / \Delta t \\ 0 & \sigma_2^2 & \sigma_2^2 / \Delta t \\ -\sigma_1^2 / \Delta t & \sigma_2^2 / \Delta t & \sigma_1^2 + \sigma_2^2 / \Delta t^2 \end{bmatrix}$$

$$= \begin{bmatrix} \sigma_1^2 & 0 & -\sigma_1^2 / \Delta t \\ 0 & \sigma_2^2 & \sigma_2^2 / \Delta t \\ -\sigma_1^2 / \Delta t & \sigma_2^2 / \Delta t & \sigma_1^2 + \sigma_2^2 / \Delta t^2 \end{bmatrix}$$